

Beyond paper counts: understanding AI mathematics

A clearer account of what is difficult, what is new and what still needs to be checked.

Alejandro Zarzuelo Urdiales | Creator and author of OPDP | 9 October 2026

722

manuscripts assessed

372

result families

17

mathematical disciplines

Hundreds of AI-generated mathematics papers raise a fundamental question: **what counts as genuine progress?** A paper count cannot distinguish a general theorem from a special case, an alternative proof or a claim that still needs checking.

Difficulty has more than one dimension

OPDP, the Open Problem Difficulty Profile, was created and authored by Alejandro Zarzuelo Urdiales. Developed with ChatGPT 5.6 Sol assistance, and further refined with newer and diverse models to capture the multidimensional complexity of mathematics, it separates the mathematical challenge from the knowledge needed, prospects for partial progress, difficulty for AI systems, and the work of checking and formally encoding a result. Public explanations make the reasons behind each assessment visible, rather than hiding them inside a single label.

From a catalogue to an evidence trail

The analysis applies OPDP to **722 manuscripts** from OpenAI's mathematics release, representing **372 result families across 17 disciplines**. It compares their stated targets with **102,819 archived problem records** and identifies **59 same-target scope matches** - connections to archived questions, not independently verified resolutions. Every manuscript receives a description, an archive comparison, a justified difficulty assessment and a conditional competition-scoring calculation. A restricted result is distinguished from its broader conjecture; companion papers do not automatically receive separate awards.

Why this changes the conversation

The contribution is an inspectable framework for judging research claims at scale. It helps researchers select targets, focus scarce expert attention on proof review and formalization, and compare different mathematical claims on explicit terms. Separating the difficulty of discovery from the difficulty of checking a result makes different research bottlenecks visible. The accompanying open data lets others inspect and revise the judgments as better evidence becomes available.

Evidence boundary: Difficulty assessments remain provisional. The report does not independently verify the proofs.

[Technical overview](#) | [Full analysis \(254 pages\)](#) | [Open data and methodology](#)

<https://github.com/alejandrozu/ulam-opdp-difficulty-atlas>